

2002text2

英文主旨： This text explores the development and limitations of robotics, tracing humanity's long history of creating tools to handle difficult work, the proliferation of intelligent devices in modern life, and the current challenges in endowing robots with true artificial intelligence, especially the gap between human cognitive abilities and machine capabilities.

中文主旨： 本文探讨了机器人技术的发展与局限，追溯了人类为应对艰难工作而发明工具的悠久历史，介绍了现代生活中智能设备的普及，并指出当前赋予机器人真正人工智能所面临的挑战，尤其是人类认知能力与机器能力之间的巨大差距。

段落1

逐句翻译与解析

1. **原文：** Since the dawn of human ingenuity, people have devised ever more cunning tools to cope with work that is dangerous, boring, burdensome, or just plain nasty.

翻译： 自从人类智慧初现以来，人们就设计出越来越精巧的工具，来应对那些危险、枯燥、繁重，或是纯粹令人不快的工作。

常考单词/短语：

- dawn n. 开端，萌芽；黎明
- ingenuity n. 创造力，聪明才智
- devise v. 设计，发明
- cunning adj. 精巧的，巧妙的（此处非贬义“狡猾的”）
- cope with 应对，处理
- burdensome adj. 繁重的，难以负担的
- plain nasty 极其糟糕的，令人不快的

长难句分析： 本句主干为 `people have devised ever more cunning tools`，`Since the dawn of human ingenuity` 为时间状语，`to cope with work...` 为目的状语，其中 `that is dangerous, boring, burdensome, or just plain nasty` 是定语从句，修饰先行词 `work`。

2. **原文：** That compulsion has resulted in robotics—the science of conferring various human capabilities on machines.

翻译： 这种冲动催生了机器人学——一门将人类的各种能力赋予机器的科学。

常考单词/短语：

- compulsion n. 强烈的冲动，欲望
- result in 导致，催生
- robotics n. 机器人学（常考学科名词）
- confer sth. on sb./sth. 将.....赋予/授予.....

3. **原文：** And if scientists have yet to create the mechanical version of science fiction, they have begun to come close.

翻译： 尽管科学家们尚未创造出科幻小说中的机械版（智能机器人），但他们已经开始接近这一目标。

常考单词/短语：

- have yet to do sth. 尚未做某事
- mechanical adj. 机械的
- come close 接近，靠近

段落1概括

本段追溯了机器人技术的起源：人类为应对艰难工作而发明工具的需求催生了机器人学，虽然尚未实现科幻作品中的完美机器人，但人类已在该领域取得了进展。

段落2

逐句翻译与解析

1. **原文**：As a result, the modern world is increasingly populated by intelligent gizmos whose presence we barely notice but whose universal existence has removed much human labor.

翻译：因此，现代世界正日益被智能装置所充斥，我们几乎注意不到它们的存在，但它们的普遍存在已经减轻了大量人类劳动。

常考单词/短语：

- as a result 因此，结果
- populate v. 充斥，居住于
- gizmo n. 小装置，小发明（考研常考熟词僻义，此处指智能设备）
- barely adv. 几乎不，勉强
- universal adj. 普遍的，广泛的
- remove v. 消除，减少（此处非“移除”）

长难句分析：本句主干为 the modern world is increasingly populated by intelligent gizmos, whose presence we barely notice but whose universal existence has removed much human labor 是两个并列的定语从句，共同修饰先行词 intelligent gizmos, 由 but 连接。

2. **原文**：Our factories hum to the rhythm of robot assembly arms.

翻译：我们的工厂随着机器人装配臂的节奏嗡嗡作响。

常考单词/短语：

- hum v. 发出嗡嗡声
- rhythm n. 节奏
- assembly arm 装配臂，机械臂

3. **原文**：Our banking is done at automated teller terminals that thank us with mechanical politeness for the transaction.

翻译：我们的银行业务在自动柜员终端上完成，这些终端会以机械般的礼貌为交易向我们致谢。

常考单词/短语：

- automated teller terminal 自动柜员终端（即ATM机，考研常考短语）
- mechanical politeness 机械的礼貌（指程序化的回应）
- transaction n. 交易，业务

4. **原文**：Our subway trains are controlled by tireless robot-drivers.

翻译：我们的地铁列车由不知疲倦的机器人司机操控。

常考单词/短语：

- tireless adj. 不知疲倦的，孜孜不倦的

5. **原文**：And thanks to the continual miniaturization of electronics and micro-mechanics, there are already robot systems that can perform some kinds of brain and bone surgery with submillimeter accuracy—far greater precision than highly skilled physicians can achieve with their hands alone.

翻译：而且，得益于电子设备和微机械技术的不断微型化，现在已经出现了能够以亚毫米级精度执行某些脑部和骨科手术的机器人系统——其精度远高于技术娴熟的医生仅凭双手所能达到的水平。

常考单词/短语：

- thanks to 多亏，由于
- continual adj. 持续的，不断的
- miniaturization n. 微型化（常考科技类词汇）

- micro-mechanics n. 微机械学
- perform surgery 执行手术
- submillimeter accuracy 亚毫米级精度
- precision n. 精度, 精确度
- highly skilled 技术娴熟的

长难句分析: 本句主干为 *there are already robot systems*, *thanks to the continual miniaturization of electronics and micro-mechanics* 为原因状语, *that can perform some kinds of brain and bone surgery with submillimeter accuracy* 是定语从句修饰 *robot systems*, 破折号后的 *far greater precision than highly skilled physicians can achieve with their hands alone* 是对 *submillimeter accuracy* 的补充说明, 构成比较状语从句。

段落2概括

本段举例说明了现代社会中智能设备的广泛应用: 从工厂机械臂、银行ATM、地铁机器人司机, 到高精度医疗手术机器人, 这些设备已大量替代人类劳动, 展现了机器人技术的实用成果。

段落3

逐句翻译与解析

1. **原文:** But if robots are to reach the next stage of laborsaving utility, they will have to operate with less human supervision and be able to make at least a few decisions for themselves—goals that pose a real challenge.

翻译: 但如果机器人要进入节省劳力的下一个阶段, 它们必须能够在更少的人工监督下运行, 并且至少能够独立做出一些决定——这些目标构成了真正的挑战。

常考单词/短语:

- laborsaving adj. 节省劳力的
- utility n. 效用, 用途
- supervision n. 监督, 管理
- pose a challenge 构成挑战, 带来挑战

长难句分析: 本句主干为 *they will have to operate... and be able to make...*, *if robots are to reach the next stage of laborsaving utility* 是条件状语从句, 破折号后的 *goals that pose a real challenge* 是对前面两个目标的同位语补充, 其中 *that pose a real challenge* 是定语从句修饰 *goals*。

2. **原文:** “While we know how to tell a robot to handle a specific error,” says Dave Lavery, manager of a robotics program at NASA, “we can’t yet give a robot enough ‘common sense’ to reliably interact with a dynamic world.”

翻译: 美国国家航空航天局 (NASA) 机器人项目主管戴夫·拉弗里说: “虽然我们知道如何让机器人处理特定的错误, 但我们还无法赋予机器人足够的‘常识’, 让它们可靠地与动态变化的世界互动。”

常考单词/短语:

- handle v. 处理, 应对
- specific adj. 特定的, 具体的
- common sense 常识
- reliably adv. 可靠地
- interact with 与.....互动/交互
- dynamic adj. 动态的, 不断变化的

长难句分析: 本句为直接引语的倒装结构, 主干为 *Dave Lavery says "..."*, 其中引语部分由 *while* 引导让步状语从句 *while we know how to tell a robot to handle a specific error*, 主句为 *we can’t yet give a robot enough ‘common sense’ to reliably interact with a dynamic world*; *manager of a robotics program at NASA* 是 *Dave Lavery* 的同位语, 说明其身份。

段落3概括

本段指出机器人技术进一步发展的瓶颈：要实现更高层次的自动化，机器人需要减少人工监督并自主决策，但目前人类还无法赋予机器人足够的常识以应对动态的现实世界。

段落4

逐句翻译与解析

1. **原文**：Indeed the quest for true artificial intelligence has produced very mixed results.

翻译：事实上，对真正人工智能的追求产生了喜忧参半的结果。

常考单词/短语：

- quest for 追求，探索
- artificial intelligence 人工智能（AI，常考缩写）
- mixed results 喜忧参半的结果，好坏参半的结果

2. **原文**：Despite a spell of initial optimism in the 1960s and 1970s when it appeared that transistor circuits and microprocessors might be able to copy the action of the human brain by the year 2010, researchers lately have begun to extend that forecast by decades if not centuries.

翻译：尽管在20世纪60和70年代曾有过一段初期的乐观时期，当时人们认为晶体管电路和微处理器可能在2010年之前就能复制人类大脑的活动，但研究人员最近已开始将这一预测延后了即便不是几个世纪，至少也是几十年。

常考单词/短语：

- despite prep. 尽管，不管
- a spell of 一段.....的时间
- initial adj. 最初的，初期的
- optimism n. 乐观，乐观主义
- transistor circuit 晶体管电路
- microprocessor n. 微处理器
- copy the action of the human brain 复制人类大脑的活动
- extend the forecast 延后预测，延长预测时间
- if not centuries 即便不是几个世纪（常考插入语，表让步）

长难句分析：本句主干为 researchers lately have begun to extend that forecast by decades if not centuries，Despite a spell of initial optimism in the 1960s and 1970s... 是介词短语引导的让步状语，其中 when it appeared that transistor circuits and microprocessors might be able to copy the action of the human brain by the year 2010 是定语从句修饰 the 1960s and 1970s，该定语从句中又包含 it appeared that... 的主语从句结构。

段落4概括

本段介绍了人工智能研究的进展与挫折：早期人们曾乐观地认为到2010年机器就能模拟人脑，但现实是这一目标被大幅延后，人工智能的发展远不如预期顺利。

段落5

逐句翻译与解析

1. **原文**: What they found, in attempting to model thought, is that the human brain's roughly one hundred billion nerve cells are much more talented—and human perception far more complicated—than previously imagined.

翻译: 在尝试模拟人类思维的过程中, 他们发现, 人类大脑中约一千亿个神经细胞比之前想象的要能干得多, 而人类的感知能力也远比想象中复杂得多。

常考单词/短语:

- model thought 模拟思维, 构建思维模型
- roughly adv. 大约, 大致
- nerve cell 神经细胞
- talented adj. 有能力的, 能干的 (此处非“有天赋的”)
- perception n. 感知, 知觉
- complicated adj. 复杂的
- previously imagined 之前想象的

长难句分析: 本句主干为 what they found... is that..., 其中 what they found 是主语从句, in attempting to model thought 是插入语作状语, that the human brain's roughly one hundred billion nerve cells are much more talented—and human perception far more complicated—than previously imagined 是表语从句; 破折号间的 and human perception far more complicated 是并列省略结构, 完整形式为 and human perception is far more complicated。

2. **原文**: They have built robots that can recognize the error of a machine panel by a fraction of a millimeter in a controlled factory environment.

翻译: 他们制造的机器人能够在受控的工厂环境中, 识别出机器面板上不足一毫米的误差。

常考单词/短语:

- recognize v. 识别, 辨认
- machine panel 机器面板
- a fraction of a millimeter 几分之一毫米, 不足一毫米
- controlled environment 受控环境, 可控环境

3. **原文**: But the human mind can glimpse a rapidly changing scene and immediately disregard the 98 percent that is irrelevant, instantaneously focusing on the monkey at the side of a winding forest road or the single suspicious face in a big crowd.

翻译: 但人类的大脑只需扫一眼快速变化的场景, 就能立即忽略其中98%无关的信息, 瞬间聚焦在蜿蜒林间小路旁的猴子, 或是茫茫人海中一张可疑的面孔上。

常考单词/短语:

- glimpse v. 瞥一眼, 扫视
- rapidly changing 快速变化的
- disregard v. 忽略, 无视
- irrelevant adj. 无关的, 不相关的
- instantaneously adv. 瞬间地, 即刻地
- focus on 聚焦于, 集中注意力于
- winding adj. 蜿蜒的, 曲折的
- suspicious adj. 可疑的, 令人怀疑的

长难句分析: 本句主干为 the human mind can glimpse... and immediately disregard..., instantaneously focusing on..., 其中 that is irrelevant 是定语从句修饰先行词 the 98 percent, instantaneously focusing on... 是现在分词短语作伴随状语, 补充说明主句动作的结果。

4. **原文**: The most advanced computer systems on Earth can't approach that kind of ability, and neuroscientists still don't know quite how we do it.

翻译: 地球上最先进的计算机系统也无法企及这种能力, 而神经科学家们至今仍不完全清楚人类是如何做到这一点的。

常考单词/短语:

- advanced adj. 先进的, 高级的
- approach v. 接近, 企及 (此处非“靠近”)
- neuroscientist n. 神经科学家
- don't know quite 不完全清楚, 不甚了解

段落5概括

本段揭示了人类认知能力的复杂性远超机器: 机器人只能在受控环境中处理精准的机械任务, 但人类大脑能快速筛选信息、聚焦关键内容, 这种能力是目前最先进的计算机系统也无法实现的, 其原理仍未被人类完全理解。

完整原文

Text 2

1. Since the dawn of human ingenuity, people have devised ever more cunning tools to cope with work that is dangerous, boring, burdensome, or just plain nasty. That compulsion has resulted in robotics—the science of conferring various human capabilities on machines. And if scientists have yet to create the mechanical version of science fiction, they have begun to come close.
2. As a result, the modern world is increasingly populated by intelligent gizmos whose presence we barely notice but whose universal existence has removed much human labor. Our factories hum to the rhythm of robot assembly arms. Our banking is done at automated teller terminals that thank us with mechanical politeness for the transaction. Our subway trains are controlled by tireless robot-drivers. And thanks to the continual miniaturization of electronics and micro-mechanics, there are already robot systems that can perform some kinds of brain and bone surgery with submillimeter accuracy—far greater precision than highly skilled physicians can achieve with their hands alone.
3. But if robots are to reach the next stage of laborsaving utility, they will have to operate with less human supervision and be able to make at least a few decisions for themselves—goals that pose a real challenge. “While we know how to tell a robot to handle a specific error,” says Dave Lavery, manager of a robotics program at NASA, “we can’t yet give a robot enough ‘common sense’ to reliably interact with a dynamic world.”
4. Indeed the quest for true artificial intelligence has produced very mixed results. Despite a spell of initial optimism in the 1960s and 1970s when it appeared that transistor circuits and microprocessors might be able to copy the action of the human brain by the year 2010, researchers lately have begun to extend that forecast by decades if not centuries.
5. What they found, in attempting to model thought, is that the human brain's roughly one hundred billion nerve cells are much more talented—and human perception far more complicated—than previously imagined. They have built robots that can recognize the error of a machine panel by a fraction of a millimeter in a controlled factory environment. But the human mind can glimpse a rapidly changing scene and immediately disregard the 98 percent that is irrelevant, instantaneously focusing on the monkey at the side of a winding forest road or the single suspicious face in a big crowd. The most advanced computer systems on Earth can't approach that kind of ability, and neuroscientists still don't know quite how we do it.

完整译文

Text 2

1. 自从人类智慧初现以来，人们就设计出越来越精巧的工具，来应对那些危险、枯燥、繁重，或是纯粹令人不快的工作。这种冲动催生了机器人学——一门将人类的各种能力赋予机器的科学。尽管科学家们尚未创造出科幻小说中的机械版智能机器人，但他们已经开始接近这一目标。
2. 因此，现代世界正日益被智能装置所充斥，我们几乎注意不到它们的存在，但它们的普遍存在已经减轻了大量人类劳动。我们的工厂随着机器人装配臂的节奏嗡嗡作响。我们的银行业务在自动柜员终端上完成，这些终端会以机械般的礼貌为交易向我们致谢。我们的地铁列车由不知疲倦的机器人司机操控。而且，得益于电子设备和微机械技术的不断微型化，现在已经出现了能够以亚毫米级精度执行某些脑部和骨科手术的机器人系统——其精度远高于技术娴熟的医生仅凭双手所能达到的水平。
3. 但如果机器人要进入节省劳力的下一个阶段，它们必须能够在更少的人工监督下运行，并且至少能够独立做出一些决定——这些目标构成了真正的挑战。美国国家航空航天局（NASA）机器人项目主管戴夫·拉弗里说：“虽然我们知道如何让机器人处理特定的错误，但我们还无法赋予机器人足够的‘常识’，让它们可靠地与动态变化的世界互动。”
4. 事实上，对真正人工智能的追求产生了喜忧参半的结果。尽管在20世纪60和70年代曾有过一段初期的乐观时期，当时人们认为晶体管电路和微处理器可能在2010年之前就能复制人类大脑的活动，但研究人员最近已开始将这一预测延后了即便不是几个世纪，至少也是几十年。
5. 在尝试模拟人类思维的过程中，他们发现，人类大脑中约一千亿个神经细胞比之前想象的要能干得多，而人类的感知能力也远比想象中复杂得多。他们制造的机器人能够在受控的工厂环境中，识别出机器面板上不足一毫米的误差。但人类的大脑只需扫一眼快速变化的场景，就能立即忽略其中98%无关的信息，瞬间聚焦在蜿蜒林间小路旁的猴子，或是茫茫人海中一张可疑的面孔上。地球上最先进的计算机系统也无法企及这种能力，而神经科学家们至今仍不完全清楚人类是如何做到这一点的。